

Institutional structures, digital inequality, and AI integration in higher education

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ABSTRACT

The global diffusion of artificial intelligence (AI) in higher education raises questions about institutional equity, as adoption patterns shaped by structural factors risk reproducing existing inequalities. This study provides one of the first systematic comparative assessments of AI integration across Nigerian university types, examining how governance structures, resources, and inter-organizational networks shape technology adoption in developing country higher education. This study conducted a comparative content analysis of 45 Nigerian universities (15 Federal, 15 State, 15 Private), assessing AI integration across six dimensions: infrastructure, curriculum, research, industry partnerships, international collaborations, and policy frameworks using structured coding scales with anchored descriptors. Drawing on an integrated framework combining institutional theory, resource-based view, and network theory, data were analyzed using one-way ANOVA, Pearson correlation analysis, and multiple regression. Results reveal moderate overall AI integration ($M = 4.79$, $SD = 1.71$) with meaningful variation across dimensions and institution types. Federal universities demonstrated the highest integration ($M = 5.25$), private universities recorded their strongest performance in curriculum integration ($M = 5.87$), and state universities showed emerging strength in policy frameworks. Institution age ($\beta = 0.43$, $p = 0.016$) and geographic location ($\beta = 0.31$, $p = 0.029$) significantly predicted integration levels, while institution type was not significant when controlling for other factors. The strong correlation between international collaborations and industry partnerships ($r = 0.74$, $p < 0.001$) suggests network positions create compounding advantages for well-connected institutions. These findings challenge assumptions that governance structure determines integration capacity and reveal how historical and geographic inequalities perpetuate through technology adoption, with implications for equitable AI deployment and international partnership design in developing country higher education.

1. Introduction

Artificial intelligence (AI) technologies are rapidly reshaping higher education globally, yet this transformation is fundamentally uneven. While institutions in well-resourced settings deploy adaptive learning systems, AI-driven research platforms, and intelligent administrative tools, universities in developing countries face structural barriers that risk reproducing and deepening existing inequalities in knowledge production and human capital development [1,2]. This unevenness raises critical societal questions: Who benefits from AI in education?

Which institutions are positioned to participate in an AI-transformed knowledge economy? And how do structural factors within societies, not merely within individual institutions, shape the distribution of AI capabilities? Understanding these dynamics is essential for educational planning and for addressing broader patterns of digital inequality that affect economic development, workforce readiness, and global knowledge equity. Systematic assessment of how institutions in diverse societal contexts integrate emerging technologies is therefore a foundational task for technology management and for informed policy responses to the social implications of rapid technological change.

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In this study, AI refers to computational systems that perform tasks ordinarily associated with human intelligence, including machine learning algorithms, natural language processing and large language models (LLMs), intelligent tutoring systems, robotics, and data analytics platforms. The assessment captures institutional engagement with this family of technologies as a whole rather than any single application, reflecting how universities present AI activity across teaching, research, and operations. These technologies can be leveraged by faculty for research and instruction, by administrators for institutional operations, and by students for learning support [3]. The emergence of generative AI tools since 2022 has particularly accelerated attention to how tertiary institutions prepare graduates for an AI-transformed workforce, with recent analyses indicating that AI competencies are increasingly essential for employability across sectors [4]. The pace of this transformation means that institutions unable to integrate AI risk producing graduates who are structurally disadvantaged in an increasingly AI-mediated labor market.

International partnerships have emerged as a primary mechanism for building AI capacity in developing regions through knowledge exchange, joint research initiatives, curriculum development support, and infrastructure sharing. The University of Michigan's \$6 million program centered on AI in Science African Faculty Fellows builds research capacity through extended faculty exchanges lasting up to 24 months [5]. Additionally, Carnegie Mellon University has established an Africa campus located in Rwanda that offers graduate programs focused on AI for social good applications [6]. These initiatives, alongside partnerships involving European, Asian, and other African institutions, suggest movement toward collaborative approaches emphasizing mutual benefit and local ownership, though the extent to which these goals are achieved varies considerably across programs. These partnerships also face sustainability challenges with societal consequences, as the 2025 USAID funding freeze for higher education partnerships in Africa made plain [7,8]. Designing durable models requires understanding how institutional contexts shape AI integration and partnership readiness in developing countries, yet this evidence base remains notably absent from the literature.

Nigeria presents a particularly compelling case for examining how structural and institutional factors shape AI integration patterns. As Africa's most populous nation with over 200 million people and the continent's largest economy, Nigeria's higher education development carries significant regional implications [9]. The Nigerian higher education landscape encompasses approximately 300 universities across three distinct governance structures: Federal universities ($n = 72$) funded by the national government, State universities ($n = 67$) supported by individual state governments across Nigeria's 36 states and Federal Capital Territory, and Private universities ($n = 160$) operated by religious organizations, foundations, or individuals [9]. This tripartite system, which parallels tiered governance structures found in other developing countries such as India, Brazil, and Kenya, creates natural variation in resources, capabilities, and governance approaches that enables systematic comparison of how different organizational and societal factors influence AI adoption patterns.

Given the recency of generative AI tools and the rapid pace of AI development, understanding of how developing country universities are incorporating these technologies remains limited. Recent comprehensive reviews of computing education in African countries have documented substantial bodies of work but emphasize the critical need for contextualized approaches that account for local realities [10]. Research on aligning African tertiary computer science education with global standards highlights both progress and persistent gaps in preparing graduates for an AI-transformed workforce [11]. A systematic review of AI in African education over the past five years identifies key trends, opportunities, and challenges while calling for more empirical research on institutional adoption patterns [12]. Despite this growing scholarly attention, no systematic comparative research has analyzed AI integration patterns across institutional types within a single national system,

limiting the ability of policymakers and international partners to develop evidence-informed strategies that account for structural variation.

This study addresses this gap through a systematic comparative assessment of AI integration across Nigerian university types. To the authors' knowledge, it is among the first to analyze AI integration patterns comparatively across institutional categories within a single national higher education system, using Nigeria's institutionally diverse system as a lens for examining how governance structures, organizational resources, and network relationships interact to shape technology adoption in developing country higher education. Drawing on a conceptual framework combining institutional theory, resource-based view, and network theory, this study assesses integration across six dimensions: infrastructure, curriculum, research initiatives, industry partnerships, international collaborations, and policy frameworks. These dimensions capture the essential organizational capabilities required for meaningful AI integration: the technical foundation (infrastructure), human capital development (curriculum and research), external knowledge networks (industry and international partnerships), and strategic direction (policy frameworks).

Consistent with a content-analytic design, this study assesses AI integration through publicly observable institutional indicators such as websites, strategic documents, course information, and partnership announcements, rather than through direct measures of classroom practice or system-level usage. These indicators capture the strategic, structural, and outward-facing dimensions of institutional AI engagement, which are themselves consequential for resource signaling and partnership readiness. The interpretive boundaries of this approach are addressed in Sections 4 and 7. Accordingly, this study aims to achieve three specific objectives. The first is to quantify the current level of AI integration across 45 Nigerian universities along the six dimensions defined above, and to compare integration across Federal, State, and Private institution types. The second is to identify the institutional and contextual factors, namely institution age, geographic location, and governance type, that predict overall AI integration. The third is to translate these patterns into evidence-informed implications for equitable and sustainable institutional capacity building and for international partnership design. These objectives speak directly to the concerns of this special issue, linking measurable institutional capacity to questions of equity and to the kind of evidence base required for sustainable technology integration in African higher education. The objectives are operationalized through two research questions:

RQ1: What is the current state of AI technology integration across Nigerian universities, and how does integration vary by institution type and dimension?

RQ2: What institutional and contextual factors predict AI integration levels in Nigerian universities?

By examining these questions, this study makes three contributions. First, it establishes a systematic baseline of AI integration across Nigerian university types, providing data valuable for Nigerian institutions, policymakers, potential international partners from any country, and the broader research community studying technology adoption and digital inequality in developing contexts. Second, it identifies institutional and contextual factors that predict integration levels, revealing how historical and geographic structures shape the distribution of AI capabilities across a national higher education system and advancing theoretical understanding of technology adoption in resource-constrained settings. Third, building on these empirical findings, it offers evidence-informed insights for equitable partnership development and policy interventions aligned with differentiated institutional capacities, contributing to the broader societal challenge of ensuring that AI's transformative potential in education does not remain concentrated in already-privileged institutions and regions.

2. Literature review

The integration of AI technologies into higher education represents a transformative force reshaping pedagogical practices, research capabilities, and institutional operations globally [1,2]. This transformation manifests across multiple domains: in teaching through adaptive learning systems, intelligent tutoring, and AI-powered content generation; in administration through chatbots, enrollment analytics, and resource optimization; and in research through automated literature review, data analysis, and hypothesis generation [3,13]. However, AI adoption patterns vary significantly across institutional contexts and between developed and developing countries [14]. This literature review examines empirical evidence relevant to understanding AI integration in Nigerian universities and its implications for international educational partnerships.

2.1. AI integration in higher education: global perspectives

AI integration in higher education has accelerated dramatically in recent years, with scholarly publications on the topic increasing two to three times between 2021 and 2022 alone [1]. While AI technologies have existed for decades in forms such as robotics, natural language processing, machine learning algorithms, and expert systems, the public release of large language models such as ChatGPT in late 2022 brought AI capabilities into mainstream awareness and prompted urgent institutional responses across higher education [15–17]. This heightened attention reflects recognition that generative AI tools represent a qualitative shift in accessibility and applicability compared to earlier AI applications [1,18].

Contemporary AI applications in higher education span multiple domains. In teaching, educators employ adaptive learning systems that adjust content difficulty based on student performance, intelligent tutoring systems that provide personalized feedback, automated assessment tools that evaluate student work, and AI-powered content generators that assist with instructional material development [19–21]. Students increasingly use AI tools for research assistance, writing support, coding help, and personalized study guidance. Administrative applications include chatbots for student services, predictive analytics for enrollment management, learning analytics for early intervention with struggling students, and resource optimization systems [4,20,22,23].

These technologies are also reshaping workforce expectations for university graduates. Industry analyses indicate that employers increasingly expect graduates to demonstrate AI literacy and ability to work alongside AI systems, regardless of their field of study [24]. Microsoft's 2024 Work Trend Index found that AI skills are becoming essential across sectors, with employers seeking graduates who can leverage AI tools to enhance productivity [24]. This shift creates pressure on universities to integrate AI not only as a subject of study but as a tool embedded throughout the educational experience. However, AI integration faces substantial challenges globally. Pedagogical concerns include impacts on critical thinking development and academic integrity, particularly with generative AI tools that can produce sophisticated text [21,25,26]. Equity concerns are acute, as AI adoption risks exacerbating educational inequalities through differential access to technology and digital literacy skills [23,27]. Ethical challenges include algorithmic bias, data privacy, and governance frameworks that have not kept pace with rapid AI development [28]. Policy responses are emerging at national and international levels, with the African Union releasing its Continental Artificial Intelligence Strategy in July 2024, providing comprehensive guidance for AI development and governance across member states [29]. A recurring challenge in this literature concerns how institutional AI integration is measured. Much existing assessment relies on institutional self-report and maturity self-audits, which are vulnerable to social desirability and uneven completion, prompting calls for indicators drawn from information that institutions

already make public. Document and website content analysis offers one such approach and has been used to map institutional digitalisation across national systems [30]. Publicly visible indicators, however, raise validation concerns of their own. Institutional theory cautions that formal structures and outward representations can become decoupled from operational practice, functioning as signals of legitimacy rather than evidence of routine use [31]. Recognizing both the value and the limits of these indicators is therefore central to interpreting any web-based assessment of AI integration, a point this study addresses directly in its measurement and validity procedures and in its limitations.

2.2. Technology adoption in African higher education

Despite infrastructure challenges, AI integration in African higher education has progressed across multiple fronts. Recent studies document AI applications in educational contexts, including AI-driven career guidance systems that help students navigate employment pathways and bridge gaps in traditional career services [32]. Computing education programs across Africa have developed innovative approaches to teaching AI and machine learning, with curricula increasingly incorporating practical projects and industry-relevant skills [10]. Research on AI for educational development highlights emerging applications in areas such as personalized learning, administrative automation, and student support systems [33]. A systematic review of AI in African education over the past five years identifies growing scholarly attention to these opportunities alongside persistent implementation challenges [12].

However, significant barriers still constrain widespread AI adoption. The digital divide manifests through inconsistent electricity supply, limited internet connectivity, and inadequate computing facilities [34,35]. Many African universities operate with aging infrastructure and unreliable power supplies that impede sustained operation of AI systems [36,37]. Uneven distribution of resources between urban and rural institutions creates significant disparities in integration capacity [38]. Human capital constraints include limited faculty trained in AI technologies, insufficient professional development opportunities, and challenges retaining technically skilled personnel [39,40]. A shortage of industry internship opportunities further limits students' abilities to link theoretical knowledge to practical AI applications [41]. These barriers are not merely technical or logistical; they reflect deeper patterns of digital inequality that risk entrenching existing disparities in global knowledge production and human capital development. As Pedro et al. [14] argue, AI's transformative potential in education cannot be realized equitably without addressing foundational inequalities in infrastructure, connectivity, and institutional capacity that disproportionately affect developing regions. UNESCO's guidance on generative AI in education further underscores that the absence of national regulatory frameworks in most developing countries leaves institutions unprotected and largely unprepared to validate and govern emerging AI tools, creating risks that extend beyond individual institutions to affect entire educational systems [42]. The structural nature of these challenges means that technology adoption patterns in African higher education cannot be understood in isolation from the broader societal conditions that shape institutional capacity and constrain equitable access to AI's benefits.

Efforts to address these challenges through contextualized approaches and collaborative networks show promise. Intra-African collaborations are opening new trajectories for knowledge production and exchange among African universities, particularly within regional bodies where facilitative policy frameworks and funding mechanisms have been established [43]. The Africa Higher Education Centers of Excellence program, a World Bank-funded initiative, has established over 80 centers across 20 countries, training more than 90,000 students and fostering regional research collaboration [44]. These regional initiatives demonstrate potential pathways for building AI capacity through collaborative approaches.

2.3. The Nigerian higher education context

Nigeria's higher education system, the largest in Sub-Saharan Africa, operates through three distinct governance structures that create meaningful variation in institutional priorities, resource allocation, and operational flexibility. Federal universities receive funding from the national government primarily through the Tertiary Education Trust Fund (TETFund), which provides relatively stable resources for infrastructure development, research support, and faculty capacity building [45]. This stable funding enables long-term planning but comes with bureaucratic oversight that can constrain rapid innovation. State universities are funded by individual state governments, creating significant variation in resources depending on each state's economic conditions and political priorities; these institutions often have strong ties to regional development agendas but face funding uncertainty [46]. Private universities operate with greater autonomy in curriculum design, hiring, and organizational decision-making, enabling rapid response to market demands, though they depend heavily on tuition revenue and philanthropic support [46–48].

Within this diverse landscape, AI-related activities are emerging across all institution types. Recent studies document AI-enabled learning platforms, learning analytics systems, and AI curriculum integration in various Nigerian universities [49–51]. Yet, AI literacy among educators and students remains foundational across most institutions, with significant gaps in both awareness and practical skills [39,52]. The COVID-19 pandemic accelerated technology adoption, forcing rapid e-learning implementation while exposing significant gaps in infrastructure, digital literacy, and institutional readiness [53–55]. This experience catalyzed increased institutional attention to digital transformation but also highlighted the substantial investments required for sustainable technology integration.

2.4. International educational partnerships

International educational partnerships in higher education can be defined as sustained institutional relationships in which partner organizations with diverse expertise and resources join together to devise and execute plans for common goals, emphasizing mutual benefit, capacity development, and local ownership rather than traditional donor-recipient dynamics [56,57]. Such partnerships take various forms including faculty exchange programs, joint research initiatives, curriculum development collaborations, student mobility arrangements, and infrastructure sharing agreements. The effectiveness of these arrangements depends substantially on alignment between partnership structures and the capacities of participating institutions.

Research on international educational collaborations identifies recurring challenges. Power asymmetries between Northern and Southern partners can undermine genuine collaboration [58]. Sustainability concerns arise when partnerships depend heavily on external funding without developing local resource bases. Misaligned priorities between partner institutions and insufficient attention to local contexts can limit partnership effectiveness [57]. The 2025 USAID funding freeze for higher education partnerships in Africa starkly demonstrated these vulnerabilities, disrupting programs across the continent when a single major funding source was withdrawn [7,8]. This disruption highlighted the urgent need for partnership models that build genuine local capacity and diversify funding sources. Understanding how Federal, State, and Private universities differ in their capacities and partnership readiness would enable international partners to develop more targeted and sustainable collaboration strategies.

The governance dimension of international AI partnerships warrants particular attention. UNESCO's Recommendation on the Ethics of Artificial Intelligence, adopted by 193 member states in 2021, establishes a global normative framework addressing transparency, accountability, fairness, and human oversight of AI systems, with specific policy action areas for education and research [59]. However, translating these

international frameworks into institutional-level policies and practices remains a significant challenge, especially in developing country contexts where institutional capacity for policy development may be limited. The gap between global AI governance aspirations and institutional implementation has societal implications, as the absence of clear institutional policies governing AI use, deployment, and ethics leaves universities exposed to the risks of uncoordinated adoption [42]. Effective international partnerships, therefore, must attend not only to building technical and research capacity but also to strengthening the governance and policy infrastructure that ensures AI integration serves broad societal interests rather than reproducing existing patterns of inequality.

3. Conceptual framework

This study employs an integrated conceptual framework combining institutional theory, resource-based view (RBV), and network theory to understand AI technology integration in Nigerian universities. While classical technology adoption theories, such as diffusion of innovations and the unified theory of acceptance and use of technology, explain individual-level adoption [60,61], they inadequately capture organizational-level technology integration in resource-constrained developing country contexts. The three theoretical perspectives selected for this framework are complementary: institutional pressures create motivation for adoption, resources determine capacity to adopt, and networks provide knowledge and support for implementation.

3.1. Institutional theory

Institutional theory examines how organizational structures, norms, and legitimacy concerns influence technology adoption [62,63]. Universities face isomorphic pressures that shape technology adoption beyond purely technical considerations [62]. Coercive isomorphism arises from formal and informal pressures exerted by other organizations upon which institutions depend, including governments and resource providers. Mimetic isomorphism results from uncertainty, leading organizations to model themselves on successful peers. Normative isomorphism stems from professionalization, as members of occupational groups share common educational backgrounds and professional associations [62].

Empirical research has applied institutional theory to understand technology adoption in higher education contexts. Scholars examining organizational change in Colombian higher education institutions found that all three types of isomorphism influenced institutional responses to environmental pressures, with coercive pressures from government policies being particularly strong [64]. In the context of Nigerian higher education, institutional theory suggests that different university types may face distinct isomorphic pressures. Federal universities, as nationally-funded flagship institutions, may experience stronger normative pressures to maintain research prominence and adopt technologies that signal institutional legitimacy [65–67]. Private universities, operating in competitive enrollment markets, may face intensified mimetic pressures to adopt visible innovations that differentiate them from competitors [68–70]. State universities, dependent on state government funding, may encounter coercive pressures shaped by regional political priorities [71–73].

3.2. Resource-Based view

Resource-based view theory explains how organizational resources and capabilities influence strategic decisions including technology adoption [74,75]. Sustained competitive advantage derives from resources that are valuable, rare, inimitable, and non-substitutable, including infrastructure, human capital, and financial assets [76]. For AI integration, critical resources include computational infrastructure enabling advanced applications, faculty expertise and technical staff

maintaining AI systems, and financial capacity supporting capital investments and ongoing operations [77,78].

Empirical research has applied a resource-based view to explore technology adoption in educational contexts. Researchers examining e-learning adoption in Ghanaian higher education found that institutional resources, particularly IT infrastructure, technical support staff, and faculty development programs, significantly predicted adoption success, with resource-rich institutions demonstrating more comprehensive implementation [79]. Similarly, a study of digital transformation in European universities demonstrated that human capital resources, specifically faculty digital competencies and dedicated technical personnel, were stronger predictors of successful technology integration than financial resources alone [80]. Resource heterogeneity is particularly salient in developing country contexts where constraints create significant capability variation across institutions [81,82]. This perspective suggests that AI integration levels should correlate positively with institutional resource availability, and that resource differences across Federal, State, and Private universities should manifest in differentiated integration patterns.

3.3. Network theory

Network theory emphasizes how inter-organizational relationships facilitate knowledge transfer and resource access [83,84]. Universities' network connections, including research collaborations, industry partnerships, and professional associations, provide channels for acquiring technological knowledge and legitimizing innovations [85]. This collaborative approach is vital in the integration of new technologies like AI, where external ties can significantly influence adoption and implementation [86].

Empirical research has demonstrated the importance of network relationships for technology adoption in higher education. A study of research collaboration networks among African universities found that institutions with stronger international ties produced more research output and demonstrated greater capacity to adopt new research technologies, with network centrality predicting institutional research productivity [87]. Research on innovation adoption in Nigerian universities specifically found that institutions with established industry

partnerships were significantly more likely to implement new technologies, as these relationships provided both resources and legitimacy for innovation [88]. AI integration networks operate at multiple levels: industry partnerships provide applied knowledge, datasets, and funding; international collaborations offer access to cutting-edge research and global standards; regional networks enable knowledge sharing among institutions facing similar challenges [89,90].

3.4. Framework integration and predictions

The integration of these three perspectives provides a unified conceptual framework for understanding AI adoption in Nigerian higher education (Fig. 1). The framework conceptualizes AI integration as simultaneously influenced by institutional pressures (institutional theory), resource capabilities (RBV), and network-facilitated knowledge transfer (network theory). These complementary lenses illuminate interconnected dimensions of technology adoption that no single theory could capture independently.

The framework yields testable predictions examined in this study. First, older institutions with established legitimacy should demonstrate higher overall AI integration due to accumulated organizational capabilities and resources. Second, AI integration should correlate positively with infrastructure capacity, human capital, and financial resources, with resource differences across institution types manifesting in differentiated integration patterns. Third, universities with stronger international collaborations and industry partnerships should exhibit higher AI integration, and these network connections should be positively correlated as inter-organizational ties tend to facilitate additional connections. Fourth, integration patterns should result from interactions among institutional, resource, and network factors rather than any single factor operating independently. The six dimensions assessed in this study (infrastructure, curriculum, research, industry partnerships, international collaborations, and policy frameworks) operationalize constructs from all three theoretical perspectives.

4. Research design and methodology

This study employed a comparative content analysis methodology to

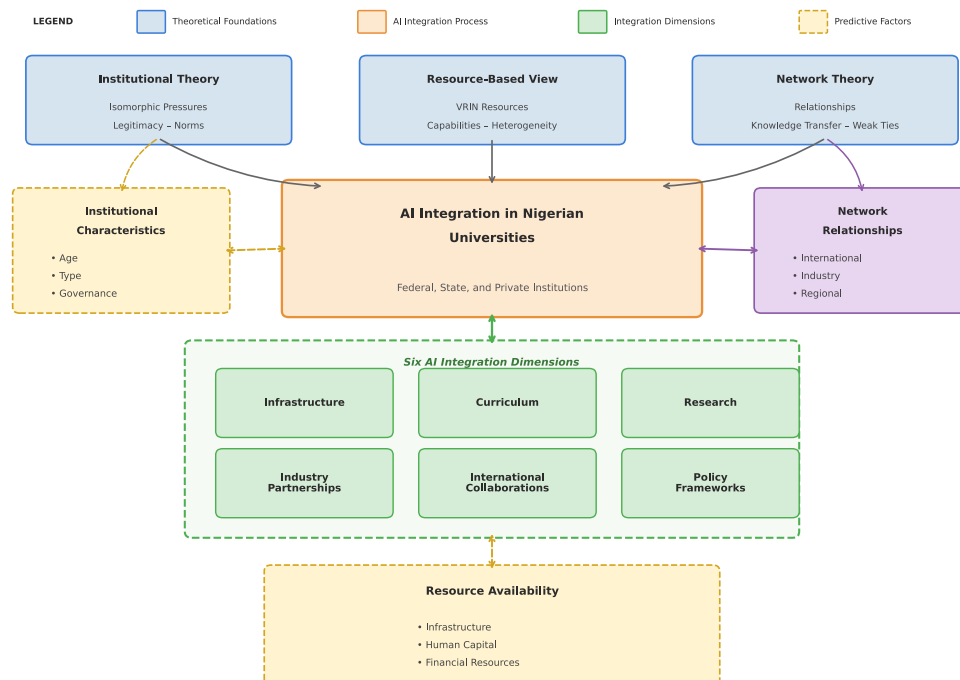


Fig. 1. Conceptual framework for AI integration in Nigerian Universities.

systematically assess AI technology integration across Nigerian universities [91]. Content analysis is well suited for examining complex organizational phenomena through publicly available documents and institutional communications [92]. This approach enabled rigorous, replicable assessment without requiring human participant data collection while generating findings directly comparable across diverse institutional contexts [93].

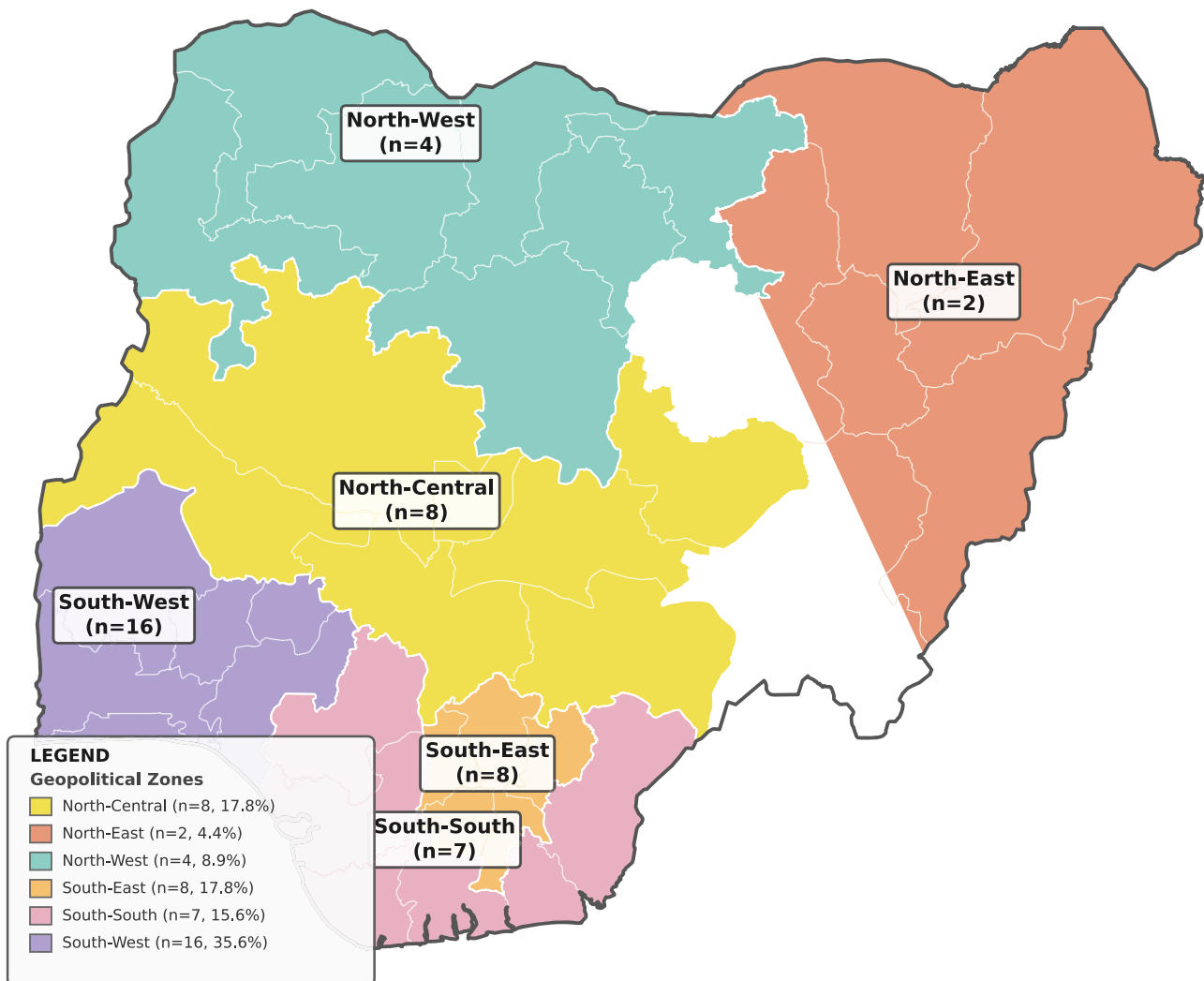
4.1. Sample selection

At sampling frame development (May 2025), Nigeria's higher education system comprised approximately 300 universities recognized by the National Universities Commission across three governance categories: Federal universities ($n = 72$, 24.1%) funded by the national government, State universities ($n = 67$, 22.4%) supported by individual state governments, and Private universities ($n = 160$, 53.5%) operated by religious organizations, foundations, or individuals [9]. A total of 45 universities were sampled (15 per stratum), a sample size determined through power analysis for pairwise between-group comparisons. With 15 institutions per group, the design achieves approximately 80% power to detect medium pairwise effect sizes (Cohen's $d = 0.50$) at $\alpha = 0.05$ [94]. For omnibus three-group comparisons, this sample provides adequate power for large effects while offering more limited sensitivity to medium effects, a constraint addressed through reporting of effect

sizes and confidence intervals alongside significance tests.

The study employed equal allocation stratified sampling (15 universities per institution type) rather than proportional allocation [95]. This design choice was methodologically grounded in three considerations. First, the study's primary purpose is comparative: the research questions explicitly focus on comparing AI integration patterns across institution types rather than estimating population-level parameters for Nigerian higher education overall. Equal sample sizes maximize statistical power for between-group comparisons and pairwise tests [96]. Second, equal allocation ensures an adequate sample size within each stratum for reliable within-group analyses, including descriptive statistics, correlation analyses, and identification of predictive factors [94, 96]. Proportional allocation would yield approximately 11 Federal, 10 State, and 24 Private universities, potentially underpowering analyses for Federal and State institutions. Third, each governance structure represents a distinct policy context worthy of independent examination; equal representation ensures that findings for smaller sectors carry equivalent analytical weight [95]. This approach aligns with established practice in comparative educational research [97].

Within each stratum, universities were selected using criteria to ensure representativeness. Geographic distribution ensured representation across all six geopolitical zones (North-Central, North-East, North-West, South-East, South-South, South-West). Fig. 2 provides a map of Nigeria's geopolitical zones with the distribution of sampled institutions.



Note: Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Fig. 2. Nigeria's geopolitical zones and sampled institutions.

Establishment period captured temporal variation, with institutions established between 1948 and 2023 categorized as Legacy (pre-1980), Expansion (1980–2010), and Recent (2011–2025). Information availability ensured only universities with accessible websites and sufficient publicly available information were included. The final sample showed Federal universities with the longest establishment history (mean = 1979), followed by State (mean = 1998) and Private (mean = 2005). Geographic distribution showed concentration in the South-West ($n = 16$, 35.6%), consistent with historical patterns, followed by South-East ($n = 8$, 17.8%), North-Central ($n = 8$, 17.8%), South-South ($n = 7$, 15.6%), North-West ($n = 4$, 8.9%), and North-East ($n = 2$, 4.4%).

4.2. Data collection

Data collection followed a systematic protocol designed to ensure consistency and comprehensiveness across all institutions. Data was collected manually between May and October 2025 by three trained researchers, following standardized procedures. Primary data sources included official university websites and affiliated departmental sites; strategic plans and institutional documents, when publicly available; course catalogs and academic program descriptions; faculty profiles and research publications; news releases and media coverage of AI-related initiatives; and partnership announcements and collaboration agreements.

Each institution was assessed through a five-step process: (1) systematic review of the main university website; (2) department-level analysis with particular emphasis on computer science, engineering, and related technical departments, as these fields serve as primary indicators of AI integration given their direct engagement with AI technologies; (3) faculty profile assessment examining individual research interests and publications; (4) review of news and events for recent AI-related announcements and activities; and (5) identification of collaboration agreements and partnerships. A standardized keyword search strategy employed primary terms (“artificial intelligence,” “AI,” “machine learning,” “data science”) and secondary terms (“robotics,” “automation,” “intelligent systems,” “deep learning”). This multi-tiered approach offered a comprehensive identification of AI-related content while minimizing false positives. To mitigate temporal inconsistency across the five-month data collection period, institutions were assessed in randomized order across strata, and a subset of early-coded institutions was re-examined at the end of the collection period to verify the stability of web content.

4.3. Coding framework

AI integration was assessed across six dimensions derived from the integrated conceptual framework described in Section 3. The dimensions operationalize constructs from all three theoretical perspectives: AI Infrastructure captures resource-based capabilities [74]; Curriculum Integration reflects institutional responses to external demands for AI competencies [62]; Research Initiatives represent accumulated human capital and knowledge production capacity [74,75]; Industry Partnerships and International Collaborations operationalize network relationships that facilitate knowledge transfer and resource access [83,85]; and Policy Frameworks capture formal organizational strategies governing technology adoption [63]. These dimensions align with categories used in prior assessments of technology integration in higher education [80,14].

A structured coding framework with 10-point anchored rating scales was developed through an iterative process. Initial dimension definitions and scoring criteria were drafted based on the theoretical framework and existing literature on technology integration assessment. These were refined through pilot coding of five institutions (not included in the final sample) to clarify anchor definitions and resolve scoring ambiguities. Each dimension uses five anchored levels: 1–2 (Minimal: no evidence or only isolated references to AI-related

activities), 3–4 (Basic: limited, foundational activities with minimal scope), 5–6 (Moderate: established activities with multiple initiatives and evident institutional commitment), 7–8 (Advanced: comprehensive activities with dedicated resources and strategic integration), and 9–10 (Cutting-edge: activities demonstrating innovation beyond standard practice with extensive scope and demonstrated excellence). Dimension-specific scoring criteria, with detailed descriptors and illustrative examples from the scored institutions, are provided in the Supplementary Dataset (Sheets 8 and 8b), enabling full replication of the coding process. An overall AI integration score was calculated as the unweighted mean of the six dimensions, treating each dimension equally as important to comprehensive AI integration.

This approach follows established practice in comparative content analysis, where coding frameworks are developed to operationalize theoretical constructs for specific research contexts rather than adopted as standardized instruments [91,92]. The quality of such frameworks is established through clarity of operational definitions, inter-rater reliability (reported in Section 4.5), and transparency of coding materials, all of which are provided.

4.4. Data analysis

Data analysis employed both descriptive and inferential statistical techniques using Python (version 3.12) with the pandas, scipy.stats, and matplotlib libraries. Descriptive statistics were calculated for all variables, including measures of central tendency and dispersion. One-way analysis of variance (ANOVA) was used to test for significant differences between institution types across all dimensions. ANOVA assumptions were assessed prior to analysis. Shapiro-Wilk tests confirmed normality of overall AI integration scores within each group (all $p > 0.05$), and Levene's test confirmed homogeneity of variance ($p = 0.517$). For dimensional analyses, 5 of 18 group-dimension combinations showed marginal normality departures; however, ANOVA is robust to moderate non-normality with equal group sizes [98–100], and Levene's tests confirmed equal variances across all dimensions (all $p > 0.05$). Because no omnibus ANOVA reached statistical significance, no Tukey HSD post hoc tests were required; for the dimension that approached significance, pairwise effect sizes (Cohen's d) were examined to gauge practical magnitude. Effect sizes were calculated using eta-squared (η^2) for ANOVA results and Cohen's d for pairwise mean differences [94]. Pearson correlation analysis examined relationships among dimensions, with a Bonferroni correction applied to control the family-wise error rate. Multiple regression analysis was conducted to identify predictors of overall AI integration. The model included Institution Age (continuous, calculated as 2025 minus the establishment year), Geographic Location (a binary variable distinguishing the South-West region from all other regions, given the historical concentration of educational resources in this region), and Institution Type (dummy-coded with Federal universities as the reference category). This specification yielded four predictors, enabling assessment of each factor's independent contribution while controlling for the others. Statistical significance was set at $\alpha = 0.05$ for all analyses.

4.5. Validity and reliability

Multiple measures were implemented to ensure validity and reliability. Inter-rater reliability was established through independent coding of 20% of institutions ($n = 9$) by two trained researchers. Cohen's kappa (κ) values ranged from 0.79 to 0.85 across dimensions, indicating substantial to almost perfect agreement [101]. Discrepancies were resolved through discussion and consensus, with coding guidelines refined accordingly. In cases where raters assigned adjacent scores and consensus favored an intermediate assessment, the midpoint was recorded (e.g., 7.5), preserving the nuance of the raters' shared judgment rather than forcing a potentially less accurate integer value. External validation cross-referenced coded information against independent

sources where available. Verification rates varied by source type: academic publications (78% of research claims corroborated), news articles (67% of major initiatives confirmed), partnership announcements (54% corroborated), and government documents (43% of policy claims confirmed through independent sources). Lower verification rates for partnerships and policy frameworks reflect limited public availability of institutional agreements and internal governance documents rather than evidence of inaccuracy; these dimensions rely more heavily on institutional websites as primary sources, a limitation inherent to the content analysis methodology. Data quality varied across institutions: 55.6% had comprehensive website information, 26.7% had limited information, and 17.8% had accessibility issues addressed through additional data collection. Complete dimensional scores for all 45 institutions are available in the Supplementary Dataset.

The six dimensions assess publicly observable, documentary indicators of AI integration rather than direct measures of classroom practice, system usage, or learning outcomes. This scope follows established content-analytic practice, in which institutional documents, websites, and public communications are treated as valid traces of organizational priorities, capabilities, and strategic positioning [30,92]. The authors acknowledge that such indicators may partly reflect institutional signaling or aspirational positioning rather than fully operational integration [31]. Three features of the design mitigate this concern without eliminating it. First, the anchored rubrics privilege concrete, verifiable evidence, such as named AI courses, identifiable research outputs, and specific partnership agreements, over general statements of intent, so that aspirational language alone could not raise a dimension score. Second, external validation cross-referenced coded claims against independent sources where available, with corroboration rates reported transparently by source type. Third, the strong internal correlations among infrastructure, curriculum, and research dimensions ($r = 0.79$ to 0.80) indicate that scores cohere as expected for a single underlying construct of institutional AI capacity, supporting convergent validity. The authors therefore interpret the scores as measures of demonstrated, outward-facing AI engagement, which is itself consequential for resource signaling and partnership readiness, while treating direct operational and pedagogical practice as beyond the evidentiary reach of this design. Triangulation with faculty and student surveys, learning-management-system analytics, or system logs would strengthen confidence in these inferences and is a priority for future research.

5. Findings

5.1. Overall AI integration patterns

The final sample of 45 universities (15 per institution type) matched the sampling design, with geographic and temporal distribution as detailed in Table 1. Addressing Research Question 1 regarding the current state of AI integration, analysis revealed (Table 2 and Fig. 3) moderate overall integration levels across the sample ($M = 4.79$, $SD = 1.71$, range: 1.83–7.83 on a 10-point scale).

Federal universities had the highest mean integration scores ($M = 5.25$, $SD = 1.64$), followed by Private universities ($M = 4.76$, $SD = 1.93$) and State universities ($M = 4.36$, $SD = 1.55$). One-way ANOVA revealed

Table 1
Sample distribution by institution type and geographic region.

	Federal	State	Private	Total	%
North-Central	3	2	3	8	17.8
North-East	0	1	1	2	4.4
North-West	2	2	0	4	8.9
South-East	3	3	2	8	17.8
South-South	2	3	2	7	15.6
South-West	5	4	7	16	35.6
Total	15	15	15	45	100.0

Table 2

Overall AI integration scores by institution type.

Institution Type	n	M	SD	Min	Max	Range
Federal	15	5.25	1.64	2.67	7.83	5.16
State	15	4.36	1.55	2.00	6.50	4.50
Private	15	4.76	1.93	1.83	7.67	5.84
Overall	45	4.79	1.71	1.83	7.83	6.00

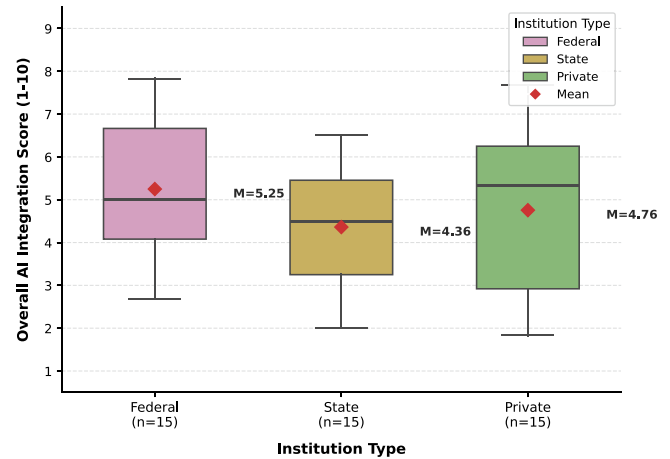


Fig. 3. Overall AI integration scores by institution type.

no statistically significant differences between institution types ($F(2,42) = 1.01$, $p = 0.372$, $\eta^2 = 0.05$), indicating that institution type accounted for little of the variation in overall integration scores.

The top five performing institutions included the University of Lagos (7.83), Afe Babalola University (7.67), Covenant University (7.33), University of Ibadan (7.25), and Ahmadu Bello University (6.83). The top performers represented both Federal universities (University of Lagos, University of Ibadan, Ahmadu Bello University) and Private universities (Afe Babalola University, Covenant University), indicating that high integration levels are achievable across governance structures. All 45 sampled institutions are listed in the Supplementary Dataset with their institution type, geographic region, and dimensional scores.

Geographic analysis revealed meaningful regional disparities in AI integration. Among regions with larger sample sizes, the South-West zone showed the highest integration ($M = 5.67$, $SD = 1.61$, $n = 16$), followed by South-South ($M = 5.19$, $SD = 1.68$, $n = 7$). The South-East region showed the lowest integration among regions with substantial samples ($M = 3.50$, $SD = 0.66$, $n = 8$). North-West universities showed the greatest within-region variability ($SD = 2.07$), reflecting heterogeneous integration patterns in that region.

5.2. Dimensional analysis

Analysis across the six AI integration dimensions revealed significant variation in institutional capabilities (Table 3 and Fig. 4). Curriculum Integration emerged as the strongest dimension ($M = 5.73$, $SD = 1.68$),

Table 3
Descriptive statistics by AI integration dimension.

Dimension	Federal		State		Private	
	M	SD	M	SD	M	SD
AI Infrastructure	4.93	1.87	3.93	1.91	4.53	2.50
Curriculum Integration	6.27	1.58	5.07	1.39	5.87	1.92
Research Initiatives	6.27	1.71	5.27	1.49	5.13	2.10
Industry Partnerships	4.90	2.17	4.53	2.20	4.60	2.35
International Collaborations	5.07	2.22	3.13	2.03	4.40	2.29
Policy Frameworks	4.07	2.09	4.20	2.04	4.00	2.00

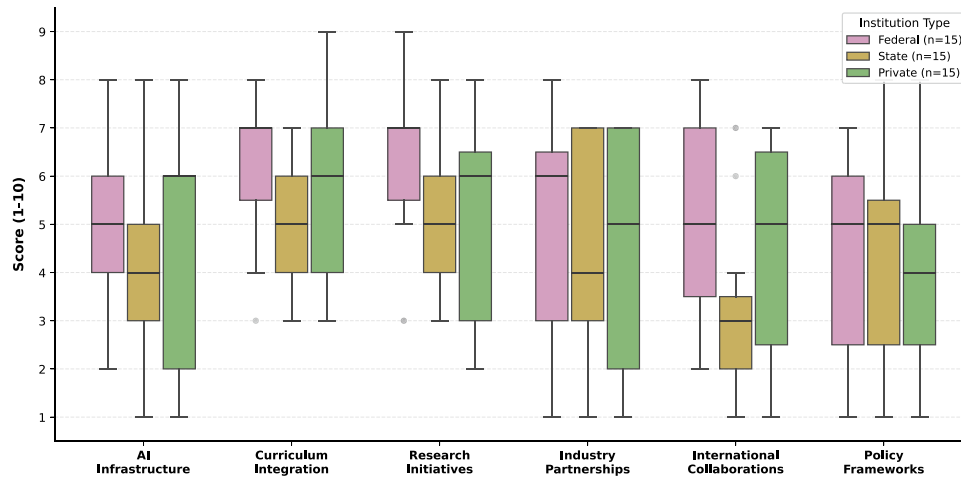


Fig. 4. AI Integration scores by dimension and institution type.

indicating that Nigerian universities have made substantial progress in incorporating AI content into academic programs. Research Initiatives demonstrated comparable strength ($M = 5.56$, $SD = 1.82$), reflecting active engagement in AI-related research.

Policy Frameworks ranked as the weakest dimension ($M = 4.09$, $SD = 2.00$), with only 27% of institutions scoring 6 or higher on formal AI strategies or governance structures. International Collaborations ($M = 4.20$, $SD = 2.28$) and Industry Partnerships ($M = 4.68$, $SD = 2.20$) also showed considerable room for improvement.

One-way ANOVA (Table 4) revealed that International Collaborations approached statistical significance ($F(2,42) = 3.03$, $p = 0.059$, $\eta^2 = 0.13$), with a medium effect size suggesting practically meaningful variation in international partnership development across institution types.

Federal universities demonstrated the strongest international collaboration scores ($M = 5.07$), followed by Private universities ($M = 4.40$) and State universities ($M = 3.13$). Post-hoc analysis revealed a large effect size for the Federal-State comparison (Cohen's $d = 0.91$), indicating that governance structure has practical relevance for international engagement capacity despite the between-group difference not reaching conventional significance thresholds.

Curriculum Integration showed no significant between-group differences ($F(2,42) = 2.07$, $p = 0.139$, $\eta^2 = 0.09$), indicating that AI curriculum development occurs across all institution types regardless of governance structure. Private universities demonstrated particular strength in this dimension ($M = 5.87$), their highest score across all six dimensions, reflecting their organizational agility in responding to market demands for AI-related programs.

5.3. Correlation analysis

Pearson correlation analysis revealed significant positive relationships among AI integration dimensions (Table 5), suggesting a hierarchical structure of institutional capabilities. The strongest relationships

Table 4

ANOVA results for AI integration dimensions by institution type.

Dimension	df	F	p	η^2
AI Infrastructure	2, 42	0.85	.434	.04
Curriculum Integration	2, 42	2.07	.139	.09
Research Initiatives	2, 42	1.81	.176	.08
Industry Partnerships	2, 42	0.11	.893	.01
International Collaborations	2, 42	3.03	.059	.13
Policy Frameworks	2, 42	0.04	.963	.00

Note. No dimension reaches $p < .05$. International Collaborations approaches significance with a medium effect size.

Table 5

Correlation matrix of AI integration dimensions.

	AII	CI	RI	IP	IC	PF
AII	—	.80***	.79***	.73***	.62***	.61***
CI	.80***	—	.79***	.56***	.58***	.43
RI	.79***	.79***	—	.74***	.65***	.61***
IP	.73***	.56***	.74***	—	.74***	.76***
IC	.62***	.58***	.65***	.74***	—	.60***
PF	.61***	.43	.61***	.76***	.60***	—

Note. AII = AI Infrastructure; CI = Curriculum Integration; RI = Research Initiatives; IP = Industry Partnerships; IC = International Collaborations; PF = Policy Frameworks. *** $p < .001$ (Bonferroni corrected). CI–PF correlation ($r = .43$) is significant at the uncorrected level ($p = .004$) but not after Bonferroni correction (adjusted $p = .053$).

were found within internal institutional functions: AI Infrastructure and Curriculum Integration ($r = 0.80$, $p < 0.001$), and between Research Initiatives and both AI Infrastructure ($r = 0.79$, $p < 0.001$) and Curriculum Integration ($r = 0.79$, $p < 0.001$). These strong correlations indicate that institutions that invest in infrastructure tend to develop corresponding curricular and research capabilities in a mutually reinforcing manner.

From the perspective of external networks, the correlation between Industry Partnerships and International Collaborations ($r = 0.74$, $p < 0.001$) was notable. While numerically lower than the internal capability correlations, it represents the strongest relationship involving external network connections. This finding suggests that globally connected institutions simultaneously develop robust local industry networks, supporting network theory predictions that connections facilitate additional connections.

Policy Frameworks demonstrated the weakest correlations with other dimensions, particularly with Curriculum Integration ($r = 0.43$, $p = 0.004$ uncorrected), the only pairwise correlation that did not survive Bonferroni correction (adjusted $p = 0.053$). This relatively weak relationship indicates a disconnect between strategic planning and operational activities, with many institutions pursuing AI curriculum development without formal institutional AI strategies to guide these efforts.

5.4. Predictors of AI integration

Addressing RQ2 (Table 6), the predictive factors analysis identified significant predictors of overall AI integration. The full model explained 30% of variance in integration scores ($R^2 = 0.30$, Adjusted $R^2 = 0.23$, $F(4,40) = 4.29$, $p = 0.006$). In social science research, R^2 values between

Table 6
Multiple regression results: predictors of overall AI integration.

Predictor	B	SE	β	t	p	95% CI for B
Institution Age	0.04	0.016	0.43	2.50	.016*	[0.01, 0.07]
Geographic Location (South-West)	1.11	0.487	0.31	2.27	.029*	[0.12, 2.09]
Institution Type	—	—	—	F(2,40) = 0.30	.744	—

Note. $R^2 = .30$, Adjusted $R^2 = .23$, $F(4, 40) = 4.29$, $p = .006$. Institution Type tested via joint F-test for State and Private dummy variables (Federal = reference). * $p < .05$.

0.10 and 0.50 are generally considered acceptable, given the complexity of human and organizational behavior [102]. The obtained R^2 of 0.30 falls within the moderate range of this benchmark, indicating that the model captures a meaningful portion of variance while acknowledging that additional unmeasured factors also influence AI integration.

Institution Age emerged as the strongest predictor ($\beta = 0.43$, $p = 0.016$), with older universities demonstrating higher integration levels. In unstandardized terms, each additional year of institutional history was associated with a 0.04 point increase on the 10-point integration scale ($B = 0.04$, 95% CI [0.01, 0.07]), strongly supporting resource-based view predictions about organizational learning and capability accumulation over time. The Geographic Location of the institution also proved significant, with South-West region universities showing higher integration scores ($\beta = 0.31$, $p = 0.029$) compared to other regions. Notably, Institution Type was not a significant predictor when controlling for other variables (joint F-test: $F(2,40) = 0.30$, $p = 0.744$), indicating that observed differences between Federal, State, and Private universities are largely mediated by factors such as institutional age, geographic location, and resource availability rather than governance structure per se. This finding has important theoretical and practical implications discussed in the following section. With 45 institutions and four predictors, the coefficients carry wide confidence intervals, so the predictive roles of institution age and geographic location are best read as associations rather than causal estimates. Their convergence with the descriptive and correlational patterns lends provisional support to the findings, while the broader interpretive limits are addressed in Section 7.3.

6. Discussion

The moderate overall AI integration ($M = 4.79$) observed across Nigerian universities reflects the complex interplay of institutional, resource, and network factors predicted by the conceptual framework. This level indicates that Nigerian universities have initiated meaningful AI integration while leaving substantial room for improvement. The empirical results provide support for all four predictions articulated in Section 3.4, though with important nuances. First, the prediction that older institutions would demonstrate higher integration was strongly confirmed: institution age emerged as the most powerful predictor of AI integration ($\beta = 0.43$, $p = 0.016$), consistent with resource-based view expectations about accumulated organizational capabilities. Second, the prediction that integration would correlate positively with institutional resources was supported by strong intercorrelations among the infrastructure, curriculum, and research dimensions ($r = 0.79$ – 0.80), indicating mutually reinforcing internal capabilities. Third, the prediction that network connections would be positively correlated and associated with higher integration was confirmed by the robust relationship between international collaborations and industry partnerships ($r = 0.74$, $p < 0.001$). Fourth, the prediction that integration patterns would reflect interactions among institutional, resource, and network factors rather than any single factor was supported by the regression finding that institution type was not a significant predictor when controlling for age

and location, demonstrating that multiple factors operate simultaneously. The following discussion interprets these findings through the three theoretical perspectives guiding this study, then examines their societal implications and transferability to other developing country contexts.

6.1. Integration patterns by institution type

Federal universities demonstrated the highest mean integration scores ($M = 5.25$), consistent with institutional theory predictions about advantages conferred by established legitimacy and stable resource bases [62,63]. These institutions, with a mean establishment year dating to 1979, have accumulated organizational capabilities over decades that facilitate technology adoption. Consistent government funding through mechanisms such as the Tertiary Education Trust Fund (TETFund) provides the financial stability necessary for sustained infrastructure investment [45]. The normative pressures associated with their role as national flagship institutions further incentivize the adoption of visible technologies like AI to maintain competitive positioning [65].

Private universities illustrated notable strength in curriculum integration ($M = 5.87$), their strongest dimension across all six areas assessed. This finding supports institutional theory predictions about mimetic pressures in competitive enrollment markets [68,69]. Private universities depend heavily on tuition revenue and must signal innovation to attract students [46]. The organizational agility afforded by more flexible governance structures enables rapid program development in response to market demands [103,104]. This pattern suggests that resource constraints do not preclude innovation when institutions possess organizational flexibility and face competitive pressures that incentivize adoption. The implication is significant: curriculum innovation can proceed even under resource constraints when institutional structures permit organizational agility.

State universities showed the lowest overall integration ($M = 4.36$), reflecting resource constraints predicted by resource-based view theory [74]. These institutions face funding variability tied to state-level economic conditions, limiting capacity for capital-intensive AI investments [105,106]. However, this finding should not be interpreted as indicating state universities lack capacity for AI development. Rather, their resource limitations may channel institutional energy toward areas requiring less capital investment, such as strategic planning and policy development. Their direct connections to state governments create unique opportunities for partnerships focused on regional development priorities.

6.2. The significance of non-significant institutional type effects

The finding that institution type was not significant when controlling for age and location (joint F-test: $F(2,40) = 0.30$, $p = 0.744$) is theoretically important and warrants careful interpretation. While descriptive statistics showed differences between Federal, State, and Private universities, these differences are largely explained by other factors. Governance structure alone does not determine integration outcomes; accumulated resources and network positions matter more. This challenges the assumption that simply identifying institutional types provides a sufficient basis for partnership design.

This finding has practical implications for international partners. Rather than seeking out partnerships based on governance category, partners should assess specific institutional profiles, including establishment history, geographic context, existing network connections, and dimensional strengths. A newer Federal university may have less integration capacity than an established Private university, despite governance-based assumptions to the contrary.

6.3. Network effects and partnership implications

The strong correlation between international collaborations and

industry partnerships ($r = 0.74$, $p < 0.001$) is a key finding with both theoretical and practical significance. Network theory predicts that inter-organizational connections facilitate knowledge transfer and resource access [85,83]. The observed correlation suggests that universities developing strong international networks are simultaneously building robust local industry relationships.

Several mechanisms may explain this pattern. International partners often facilitate local industry connections through their own networks. Globally engaged faculty attract industry attention through enhanced visibility and demonstrated competence. The capabilities developed through international collaboration, including research skills, project management experience, and exposure to global standards, enhance attractiveness to industry partners. This finding suggests that international and industry partnerships should be pursued as complementary rather than competing strategies. Effective international partnerships can serve as pathways to industry engagement, while industry connections can enhance the attractiveness of the institution to international collaborators.

The sustainability implications of this network interdependence, particularly in light of recent disruptions to international funding mechanisms, are examined further in [Section 6.5](#).

6.4. The policy framework gap

The weakness in policy frameworks across all institution types indicates a critical gap between operational AI activities and strategic institutional planning. Many universities are adopting AI technologies reactively rather than through deliberate institutional strategy. The weak relationship between policy frameworks and curriculum integration underscores this disconnect: institutions are developing AI curricula without formal policies to guide these efforts.

This gap has implications for the sustainability of partnerships. Effective international collaborations typically require clear institutional policies and governance frameworks to ensure alignment of objectives, resource allocation, and continuity beyond individual relationships. The African Union's Continental Artificial Intelligence Strategy (2024) provides a continental framework that institutions could adapt to develop institution-specific policies addressing ethical AI deployment, faculty capacity building, and alignment with national development priorities.

The regional concentration of high-performing universities in the South-West ($\beta = 0.31$, $p = 0.029$) raises equity concerns, as universities in northern and eastern regions may face structural disadvantages in AI development. Regional consortium models that enable resource sharing and knowledge transfer between well-resourced and developing institutions offer one mechanism for addressing these disparities.

6.5. Societal implications

The finding that institution age and geographic location predict AI integration more strongly than governance type reveals a pattern with significant societal implications. These predictors are not characteristics that institutions can readily change; they are structural features rooted in historical patterns of resource distribution, economic development, and educational investment. Older institutions have accumulated organizational capabilities, reputational capital, and network connections over decades that newer institutions cannot rapidly replicate, regardless of their governance structure or strategic intent. Similarly, the concentration of AI capacity in the South-West region reflects the historical concentration of economic activity, infrastructure investment, and international connectivity in the Lagos-Ibadan corridor.

This pattern mirrors broader dynamics of digital inequality documented across developing regions, where technology adoption tends to follow and reinforce pre-existing patterns of advantage [34,38]. In South Asian higher education, resource disparities between established and newer institutions similarly predict technology adoption capacity,

with institutional age and geographic proximity to economic centers shaping access to the resources required for meaningful integration [82]. In South African higher education, the legacy of historically advantaged and disadvantaged institutions continues to shape digital transformation outcomes decades after formal policy equalization [38]. The Nigerian findings thus contribute to a growing body of evidence that AI technologies, far from leveling educational playing fields, risk deepening structural inequalities when adoption depends on resources and capabilities that are themselves unequally distributed across societies.

The policy framework gap carries societal implications that extend beyond institutional management. When universities adopt AI technologies without formal strategies governing their deployment, they risk uncoordinated implementation that can amplify algorithmic biases in educational decision-making, compromise student and faculty data privacy, create inequitable access to AI tools within institutions, and produce graduates with inconsistent AI competencies [42]. The disconnect between policy development and curriculum integration is particularly concerning from a societal standpoint: it indicates that institutions are embedding AI into their academic programs without the governance infrastructure needed to ensure that this integration is ethically sound, pedagogically deliberate, and aligned with broader societal values. This gap exists at multiple levels. At the institutional level, the absence of formal AI strategies means that adoption decisions are often made by individual departments or faculty members without coordination. At the national level, the regulatory frameworks needed to guide institutional AI governance remain underdeveloped in most African countries [29]. At the global level, while UNESCO's Recommendation on the Ethics of Artificial Intelligence (2021) provides a comprehensive normative framework, translating these principles into institutional-level policies and practices represents a significant implementation challenge, particularly in developing country contexts where institutional capacity for policy development may be limited [59]. Addressing this multi-level policy gap requires coordinated action that connects global governance frameworks to institutional practice through national regulatory intermediation.

The network interdependence identified in [Section 6.3](#) carries societal implications that warrant critical examination. Because connections compound, well-connected institutions accumulate advantages while less-connected institutions face growing marginalization. Federal universities, which demonstrated the strongest international collaboration scores, are positioned to leverage these connections for further industry engagement, research funding, and institutional prestige. State universities, with significantly lower international collaboration scores, face structural barriers to entering these reinforcing networks. This network inequality raises questions about how international partnerships are designed and targeted. Partnerships that engage only already-connected institutions may strengthen those institutions while doing little to address systemic inequalities in AI capacity [58,57]. The 2025 USAID funding freeze for African higher education partnerships exposed the fragility of externally dependent partnership models and raised fundamental questions about whether such partnerships build genuine local capacity or create dependency that ultimately serves the interests of Northern partners more than Southern ones [7]. These dynamics suggest that equitable international AI partnerships require deliberate attention to inclusion of less-connected institutions, diversification of funding sources, and mechanisms that ensure partnership benefits reach beyond a small number of already-advantaged universities.

6.6. Transferability beyond Nigeria

Although this study focused on Nigerian universities, the findings have relevance for developing countries with similarly tiered higher education systems [107]. India's system of central, state, and private universities, with central universities funded by the national government and state universities dependent on individual state resources,

presents structural parallels [108] that suggest similar dynamics may operate: institution age, geographic proximity to economic centers, and network connections likely shape AI integration patterns in ways that transcend governance categories [109]. Brazil's federal, state, and private university structure and Kenya's public-private divide in higher education similarly create governance-based variation in resources, autonomy, and institutional priorities [110,111]. The finding that governance type was not a significant predictor when controlling for institutional age and geographic location may be particularly transferable to these contexts, as it challenges the common assumption that governance structure serves as a reliable proxy for institutional capacity.

The six-dimensional assessment framework developed for this study (infrastructure, curriculum, research, industry partnerships, international collaborations, and policy frameworks) and the comparative content analysis methodology offer transferable tools for assessing AI integration in other developing country contexts. The framework captures organizational capabilities grounded in established theoretical constructs: resource-based capabilities (infrastructure), institutional responses to external demands (curriculum), accumulated human capital (research), and network relationships (partnerships and collaborations). These constructs are not Nigeria-specific; they reflect universal dimensions of organizational capacity for technology integration. Researchers in other developing regions could adapt the coding rubrics and scoring scales to their contexts, enabling cross-national comparisons that would advance understanding of how diverse institutional environments shape AI adoption patterns. Such comparative studies would also allow for testing whether the predictive significance of institution age and geographic location holds across different national contexts, or whether country-specific factors moderate these relationships.

7. Implications and conclusion

Taken together, the study makes three contributions. Empirically, it provides one of the first cross-type baselines of AI integration within a single African national system, quantified along six theoretically grounded dimensions for 45 universities. Theoretically, it shows that governance type, the categorization most often used to organize partnership and policy decisions, loses explanatory power once institution age and geographic location are taken into account, redirecting attention from formal category toward accumulated capability and network position. Methodologically, it offers a transparent and replicable content-analytic instrument with anchored rubrics and reported reliability, which other researchers can adapt for cross-national comparisons. The study is timely because external funding shocks, such as the 2025 USAID freeze, have made evidence-based partner assessment an immediate practical concern rather than a long-term aspiration.

7.1. Implications for policy

The identification of institution age and geographic location as significant predictors of AI integration, while institution type was not significant when controlling for other factors, suggests that policy interventions should focus on mechanisms that build institutional capacity regardless of governance structure. Newer universities and those in underserved regions may benefit from targeted support programs that accelerate capability development. Such programs could include dedicated infrastructure grants for computing facilities and AI laboratories, faculty development initiatives providing training in AI pedagogy and research methods, mentorship arrangements pairing emerging institutions with established universities, and access to shared cloud computing resources that reduce individual infrastructure burdens. The National Information Technology Development Agency (NITDA) and TETFund could coordinate such initiatives, ensuring resources reach institutions with the greatest development potential [112]. Such coordinated efforts would help address regional disparities and accelerate AI capacity building across the Nigerian higher education sector.

The policy framework gap identified across all institution types indicates a need for sector-wide attention to strategic planning and governance. Nigerian universities would benefit from developing formal AI strategies that articulate integration goals, resource allocation priorities, and partnership approaches. For faculty, such strategies should address professional development pathways, research support mechanisms, and guidelines for AI use in teaching and assessment. For students, strategies should articulate AI literacy expectations across programs, experiential learning opportunities, and ethical frameworks for AI use in academic work. The National Universities Commission could facilitate this development through guidance documents, benchmarking exercises, and incentive mechanisms that encourage strategic planning.

The mutually reinforcing nature of partnership networks suggests that policies promoting one form of partnership may generate spillover benefits for the other [85,83]. Government programs supporting international engagement, such as faculty mobility schemes and research exchange programs, may simultaneously strengthen industry relationships. Similarly, policies encouraging university-industry collaboration may enhance international visibility and attractiveness to foreign partners. This underscores the value of integrated partnership policies that recognize the interconnected nature of institutional networks rather than treating international and industry engagement as separate domains.

7.2. Implications for international partners

The differentiated institutional profiles suggest that effective international collaboration strategies must be tailored to institutional characteristics and capabilities rather than governance structure alone. Federal universities' combination of strong research capacity ($M = 6.27$ for Research Initiatives) and international collaboration experience ($M = 5.07$ for International Collaborations) positions them well for comprehensive research partnerships that emphasize faculty exchange and joint research projects. Their established infrastructure, faculty expertise, and administrative capacity support sustained research collaboration.

Furthermore, private universities' excellence in curriculum integration ($M = 5.87$) and organizational agility positions them for innovation-focused collaborations. Their industry orientation and entrepreneurial culture align with partnership models emphasizing practical application and technology transfer. Shorter-duration, project-based partnerships that leverage private universities' flexibility may prove more effective than long-term institutional arrangements that require extensive bureaucratic navigation.

State universities, despite lower overall scores, offer unique opportunities through their direct connections to state governments. These connections create pathways for partnerships focused on public sector objectives and regional development priorities. Collaborations aligned with state development plans and addressing regional challenges may find particularly receptive partners among state universities.

7.3. Limitations and future research

Several limitations should be acknowledged. First, regarding measurement, the study relies on publicly observable documentary indicators; these may underestimate integration at institutions with limited or outdated web presence, and, as noted in Section 4.5, may capture institutional signaling alongside operational practice. The anchored rubrics, illustrative examples, and external validation rates provided in the Supplementary Dataset offer transparency on the evidentiary basis for each score. Second, regarding sampling, equal allocation across the three institution types maximizes comparative power but results in a sample that is not proportionally representative of the national system, and selecting institutions with accessible websites may modestly bias the sample toward more digitally visible universities.

Third, regarding analysis, the modest sample of 45 institutions limits power to detect small effects and the stability of regression coefficients, so estimates are reported with confidence intervals and interpreted as associations rather than causal effects. Fourth, regarding design, the cross-sectional approach captures integration at a single point in time and cannot track the rapid changes characteristic of AI adoption. Fifth, regarding scope, department-level analysis prioritized computing and engineering fields, so integration in non-technical disciplines may follow different patterns, and the focus on Nigeria limits direct generalizability to other systems despite the structural parallels discussed in Section 6.6. Future research using longitudinal designs, qualitative methods, and triangulated data sources would address several of these constraints.

7.4. Conclusion

This study established that Nigerian universities have initiated meaningful AI integration efforts, yet significant gaps remain, particularly in policy frameworks and international collaborations. More importantly, the findings challenge assumptions that governance structure alone determines integration capacity. Institutional age, geographic location, and network connections matter as much, if not more, than whether a university is Federal, State, or Private. This insight reframes how international partners should approach collaboration: not by selecting institution types, but by understanding specific institutional profiles and designing partnerships accordingly.

Institutional diversity, the very characteristic that might appear to complicate partnership design, represents a strategic asset when collaborations leverage complementary strengths across institution types. Federal universities can contribute research infrastructure and international networks, private universities can offer curriculum innovation and industry connections, and state universities can provide pathways to regional development priorities and public sector engagement. Effective international educational partnerships must move beyond traditional donor-recipient models toward collaboration that builds genuine local capacity. The findings establish essential baseline data for evidence-informed partnership development and contribute to a broader understanding of technology adoption in developing-country higher education contexts. As AI technologies continue reshaping higher education globally, ensuring that universities across all regions can meaningfully participate in this transformation is both an educational imperative and a matter of global equity.

Declaration of generative AI and AI-Assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly to check internal consistency across manuscript sections and improve language and readability. After using this tool, the authors reviewed and edited all content as needed and take full responsibility for the content of the published article.

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Ethical statement

This study analyzed publicly available information from university websites and institutional documents. No human participants were involved, and no personally identifiable data were collected. Accordingly, institutional review board approval was not required. All data sources are publicly accessible institutional communications.

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An earlier version of this manuscript was posted as a preprint on Research Square through Springer Nature's "In Review" service. The preprint appears under the title "Building Sustainable International Partnerships: Evidence-Informed Strategies Based on a Comparative Analysis of Nigerian Universities' AI Integration and Readiness." The current manuscript has been substantially revised with a new title, expanded theoretical framework, and reframed analysis emphasizing societal implications. A related conference paper drawing on the same dataset, titled "Predicting International Partnership Readiness: What Institutional Factors Shape AI Integration Capacity in Nigerian Universities' Engineering Programs?", was presented at the 2026 ASEE Annual Conference (International Division). The conference paper focuses on partnership readiness implications with a distinct framing, scope, and analytical emphasis from the present article.

CRediT authorship contribution statement

Daniel I. Adeniranye: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stephanie J. Lunn:** Writing – review & editing, Validation, Supervision, Resources. **Olatunde Mosobalaje:** Writing – review & editing, Validation, Methodology. **Precious Eze:** Validation, Methodology, Formal analysis. **Bruk Berhane:** Writing – review & editing, Supervision, Resources. **Daniel Adeniyi:** Visualization, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.caeo.2026.100400](https://doi.org/10.1016/j.caeo.2026.100400).

Data Availability

The datasets generated and analyzed during the current study, including the complete dimensional scores for all 45 institutions, scoring rubrics, and statistical outputs, are available in the Supplementary Dataset accompanying this article. All primary data were collected from publicly accessible university websites and institutional documents.

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